Exam 3: Study Guide

This study guide lists skills you need to be able to do for your exam. It does not tell you *how* to do the skills. Skills may be tested individually, together, or not at all. It is up to you to determine what skills you need already know, and which skills you need more practice on. Review relevant topics in ALEKS, problems from the end of each chapter of your textbook, worksheets, and lab experiments.

Chapter 8 - Gases

1) Remember the assumptions of the kinetic molecular theory of gases
   a) Section 8.3
2) Predict the change in a gas property (pressure, volume, temperature, amount of gas), given a scenario where some properties are changed and others held constant.
   a) Section 8.4.1-8.4.4 (Simple Gas Laws – Boyle, Charles, Guy-Lussacs, Avogadro)
3) Calculate pressure, volume, moles of gas, or temperature using the ideal gas law (PV = nRT)
   a) You may need to convert units of pressure, volume or temperature into the correct units for R. See reference sheet for conversion factors.
      i) Section 8.2 (Pressure Conversions)
      ii) Section 8.5 (Ideal Gas Law)
4) Remember Standard Temperature and Pressure Conditions (O ^{0}C and 1 atm)
   a) Utilize molar volume at STP as a conversion factor (1 mol = 22.4 L)
5) Calculate the partial pressure of a gas given information about other components in the mixture and total pressure.
   a) Section 8.6
6) Calculate the partial pressure of a gas evolved from a chemical reaction given the vapor pressure of water at that temperature and barometric pressure.
   a) (May be asked to use partial pressure to find moles of gas or mass of gas)
   b) Section 8.6

Chapter 9 – Aqueous Solutions

1) Define and differentiate between solute, solvent and solution.
    a) Section 9.1
2) Rank molecules in order of increasing (or decreasing) relative solubility based on strength of intermolecular forces between solute and solvent molecules.
    a) Section 9.1.1
3) Apply solubility rules (given) to predict if an ionic compound would be aqueous or solid in solution.
    a) Section 9.1.3
4) Recognize that an aqueous ionic compound exists as cations and anions
5) Identify solutions as being strong, weak, or non-electrolyte based on chemical composition of solute.
    a) Section 9.1.2 (very brief)
6) Differentiate between electrolyte strength, and concentration (dilute, concentrated or unsaturated/saturated.)
    a) Section 9.2 (unsaturated/saturated)
7) Describe the effects of temperature on the solubility of gases and solids in solution. For gases, describe the effects of pressure as well.
    a) Section 9.2.1 (Solids in solution)
    b) Section 9.2.2 (Gases in solution)
8) Be able to calculate the concentration of a solute in solution, and use concentration as conversion factors:
    a) Section 9.3.1
        i) Mass percent (m/m) % (g solute per 100 g solution)
        ii) Volume percent (v/v) % (mL solute per 100 mL solution)
        iii) Mass/Volume (m/v) % (g solute per 100 mL solution)
    b) Section 9.3.2
        i) Molarity (M) (mole solute per 1 L solution)
    c) Section 9.5
        i) Molality (m) (mole solute per 1 kg solvent)
9) Apply the dilution equation ($M_1V_1 = M_2V_2$) to calculate an unknown variable when a solution is being diluted
    a) Section 9.4.2
    b) Applicable for questions that explicitly say dilution, or imply a dilution using words to describe the process.
10) Rank solutions by relative freezing/boiling point.
    a) Section 9.5.1
11) Determine the direction of solvent flow given molecular views of osmosis cells.
    a) Section 9.5.2 -

Chapter 10 – Chemical Reactions

1) Identify from a submicroscopic diagram that a change is a physical change or a chemical change.
   a) Review from Chapter 4
2) Balance a chemical reaction
   a) Section 10.3
3) Translate a chemical reaction written in words to symbols into a balanced chemical equation
   a) For example: "Write the balanced chemical equation for sodium hydroxide reacting with sulfuric acid to form sodium sulfate and water"
      i) $2NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2H_2O(l)$
   b) Section 10.1, 10.2
4) Write a balanced chemical equation for a chemical reaction given sub-microscopic drawings of compounds before and after a chemical reaction.
5) Classify a reaction as one of the following types by looking at the pattern of a reaction:
   a) Section 10.4
   b) What Atoms Do:
      i) Double-displacement (Double-replacement)
      ii) Decomposition
      iii) Combination (synthesis)
      iv) Single-displacement (Single-replacement)
   c) Overall Process
      i) Acid-base (Neutralization)
      ii) Gas-forming
      iii) Precipitation
      iv) Redox (Oxidation-Reduction)
         (1) Redox with O_2 gas = Combustion
6) Predict the products of a double-displacement reactions, or apply given reaction classification to predict products of other types of reactions.
   a) Section 10.5
7) Recognize the double-displacement reactions that produce the following compounds are actually gas evolution reactions:
   a) $H_2S(g)$
   b) "H_2CO_3" actually forms $H_2O(l)$ and $CO_2(g)$
   c) "H_2SO_3" actually forms $H_2O(l)$ and $SO_2(g)$
   d) "NH_4OH" actually forms $H_2O(l)$ and $NH_3(g)$
8) Write Molecular, Complete Ionic, and Net Ionic equations for reactions that occur in aqueous solutions
   a) Section 10.6.1
9) Assign oxidation states (oxidation numbers) to elements within a compound
   a) Section 10.8.1
10) In a chemical reaction, identify which elements are reduced, and oxidized
   a) Section 10.8.2
11) Write half-reactions for simple redox reactions
   a) Section 10.8.3
   b) i.e. (some ion) + electrons → (neutral element)
12) Use the activity series of metals to predict whether a reaction will be spontaneous
   a) Section 10.9.1

13) Identify which half reaction would occur at the anode or cathode of a battery
    a) Section 10.9.2